Beyond Representation: Index Structure Limitations in Dense Retrieval

Alessandro Magnani

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Why this talk

- Dense retrieval is everywhere: e-commerce, enterprise search, RAG.
- Practical deployments rely on **approximate nearest-neighbor (ANN)** indices (HNSW, IVF) for speed.
- Most research focuses on the *representation*—can the embedding capture the right semantics?
- This talk: the index itself imposes a separate, quantifiable failure mode that disproportionately attacks a specific population of queries.

Headline finding

Even when the embedding has enough representational capacity, switching from exact (Flat) to HNSW search forfeits $\sim 63\%$ **of achievable Recall@10** on identifier queries—vs. less than 2% on standard queries.

The two camps

Sparse retrieval (BM25)

- Inverted index over tokens.
- Exact lexical matching.
- Strong on rare terms, IDs, named entities.

Dense retrieval

- Continuous embedding + nearest-neighbor search.
- Strong semantic generalization.
- Indispensable in production for fluency & multilinguality.

- Prior theory (Tay et al. 2020; Fan et al. 2022): *representation capacity* of dense models can match sparse—given enough dimensions.
- But: practical dense retrieval uses ANN, not exact search.
  What does that cost?

“Spear-fishing” queries

Definition (operational)

Queries whose relevance hinges on a single, low-frequency, high-IDF token: identifiers (SKUs, part numbers), rare terms, precise named entities.

- In production: a small fraction of total traffic.
- But operationally critical: failures yield empty search-result pages, abandoned carts, broken RAG retrievals.
- In our experiments:
  - **ID subset**: doc identifier used as query (extreme case).
  - **Rare subset**: uni/bigram terms appearing in ≤ 5 docs.
  - **Standard subset**: real user queries (control).

The mechanism: centroid dilution (theory)

IVF clusters the corpus into N cells; at query time only the closest n_{probe} centroids are searched.

Centroid of cluster C: $\quad c \;=\; \dfrac{1}{|C|}\sum_{d\in C} f(d) \;=\; \cdots + \alpha_{t^*}\, p\, \phi(t^*) + \ldots$

For a query dominated by rare token t^*:

$$\langle c, q\rangle \;\approx\; \alpha_{t^*}^2\, p \;=\; \log^2\!\left(\tfrac{M}{n_{t^*}}\right) \cdot \frac{n_{t^*,C}}{M/N}$$

- α_{t^*} is large for rare tokens (high IDF).
- But p (in-cluster prevalence) is small for rare tokens.
- JL projection adds noise floor $\epsilon \approx \sqrt{\log V/d}$.
- Rare-token centroid signal lost when $\alpha_{t^*}^2 p \lesssim \epsilon$.

Larger embedding d helps ϵ, but cannot fix small p.

Experimental setup

Datasets

- **Home Depot**: 54,667 product titles
- **MS MARCO Passage v1.1**: 82,360 passages

Three query subsets per dataset
(1,000 queries each)

- Standard | Rare (≤ 5 docs) | ID

Representation

- Char-trigram + JL projection
- $d \in \{256, 512, 1024\}$
- Unsupervised

Indices (FAISS)

- Flat (exact)
- IVF (`nlist`, `nprobe` sweep)
- HNSW (M=16, `efC`=200, `efS` sweep)
- BM25 (Pyserini)

Why unsupervised trigram, not DPR / Sentence-BERT?

A supervised encoder abstracts away the lexical detail that *defines* a spear-fishing query. The Flat baseline would already fail on IDs—conflating representation failure with indexing failure. We need a representation that *provably* preserves lexical distances (Johnson–Lindenstrauss).

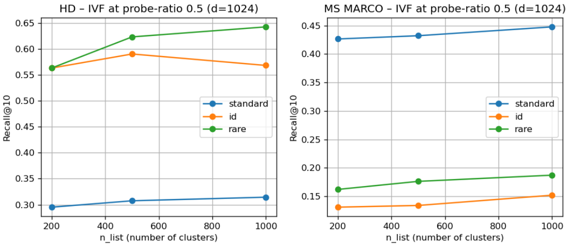
Headline result: the gap

Dataset	Subset	Flat	IVF	HNSW	BM25
HD	standard	0.327	0.314	0.321	0.447
HD	rare	0.751	0.642	0.619	0.878
HD	**id**	**0.874**	0.590	**0.327**	1.000
MS MARCO	standard	0.471	0.448	0.446	0.770
MS MARCO	rare	0.205	0.187	0.166	0.688
MS MARCO	**id**	**0.174**	0.152	**0.071**	—

Recall@10 at $d=1024$, across-token trigrams, best ANN sweep config.

- Flat→HNSW gap (relative): standard ≈ **2%** rare ≈ **18%** id ≈ 63%.
- Same monotone pattern on both datasets.
- All gaps statistically significant (paired bootstrap 95% CI excludes zero).

Indirect evidence: IVF cluster granularity



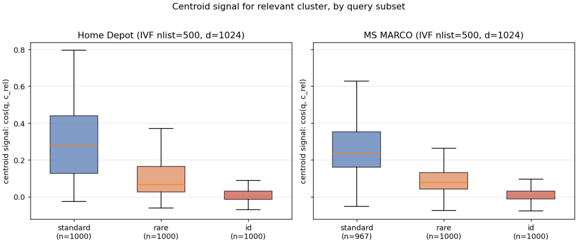
Theory predicts: smaller clusters $\Rightarrow$ less centroid dilution $\Rightarrow$ rare/ID queries benefit more from finer clustering.

Data: rare & ID gain *2–3$\times$ more* (relative) than standard going from n_{list}=200 $\rightarrow$ 1000.

Asymmetric improvement is exactly what Eq. (8) predicts.

Direct evidence: measure the centroid signal

For each query, compute the centroid signal $\langle q, c_{\text{rel}} \rangle$ on the centroid of the query's relevant cluster (IVF, n_{list}=500, d=1024).

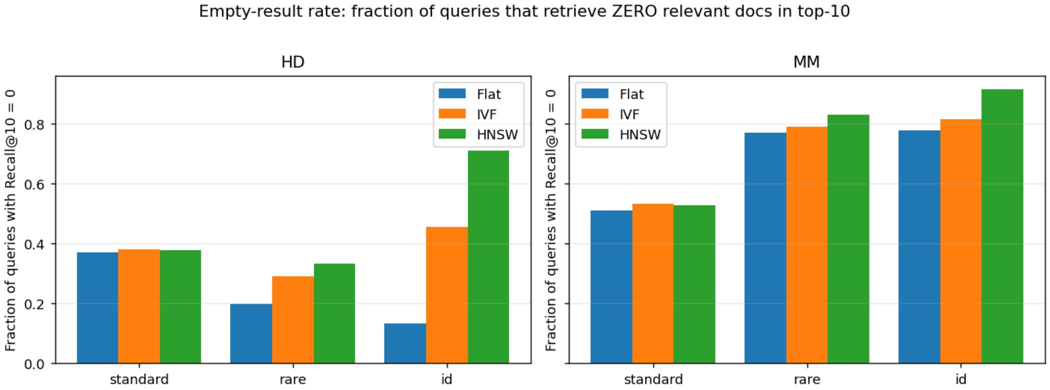


Subset (HD)	signal	rank	hit@250
standard	0.279	2	97%
rare	0.066	69	81%
id	0.007	216	59%

$\sim$ 40$\times$ collapse from standard to ID: centroid signal disappears into the JL noise floor exactly as Eq. (8) predicts.

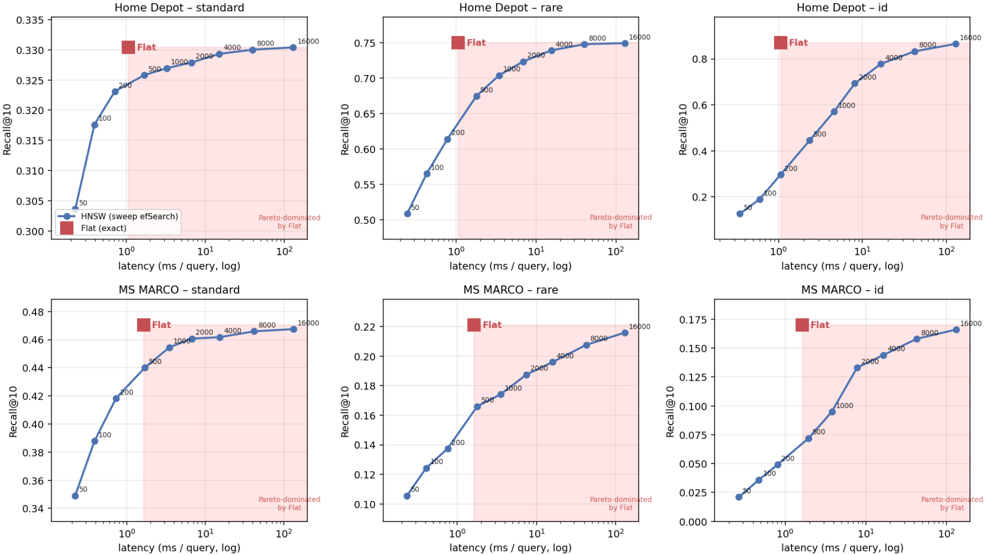
For HD-ID queries, 41% of relevant clusters fall outside the top-half by centroid score—*structurally unrecoverable*.

The empty-result rate



- HD-id Flat: **13%** of queries return nothing relevant in top-10.
- HD-id HNSW: **71%**.
- Standard subset: barely moves.
- *This is the production-visible symptom.*

The recall–latency trade-off changes shape



HNSW efSearch sweep (standard FAISS recipe M=16, efC=200) vs. Flat (red square). Red region = strictly

The cost-to-close ratio (corpus-invariant)

`efSearch` needed to reach 95% of Flat's Recall@10:

Subset	efS	ms/q	vs. Flat
HD-standard	100	0.39	**2.7$\times$ faster**
HD-rare	2,000	6.94	6.5$\times$ slower
HD-id	8,000	41.9	40$\times$ **slower**
MM-standard	1,000	3.48	2.1$\times$ slower
MM-rare	16,000	129.6	79$\times$ slower
MM-id	16,000	132.0	81$\times$ **slower**

- Standard queries: HNSW gives the canonical ANN speedup.
- ID queries: need 80–160$\times$ more efSearch budget than standard.
- Search cost scales $\sim$linearly in efSearch $\Rightarrow$ this ratio *generalizes to any corpus size*.

“But this is only because the embeddings are simple”

Reviewer question we anticipated: *do learned dense encoders avoid this?*

Re-ran the full grid with `all-MiniLM-L6-v2` (Sentence-BERT, 384-dim, supervised).

Dataset	Subset	Trigram Flat R@10	**MiniLM Flat R@10**
HD	standard	0.327	0.298
HD	rare	0.751	0.428
HD	id	0.874	**0.010**
MS MARCO	standard	0.471	**0.894**
MS MARCO	rare	0.205	0.226
MS MARCO	id	0.174	**0.001**

- MiniLM nearly doubles MS MARCO standard recall—as designed.
- MiniLM **cannot distinguish identifiers**: “12345” ≈ “67890”.
- Representation collapses *before* the index has a chance to bias.
- This is exactly why a supervised encoder is the *wrong instrument* for our study—and exactly why hybrid retrieval is needed in production.

What this means for production

Two ways to read the gap

- **Default efSearch:** HNSW forfeits $\sim 63\%$ of achievable recall on identifier queries, $\sim 18\%$ on rare, $\sim 2\%$ on standard. *Visible:* 71% of HD-ID HNSW queries return nothing relevant.
- **Match Flat recall:** on ID you must pay 40–80$\times$ Flat latency. ANN's speedup advantage *evaporates* on the population that needs reliability the most.

Recommendations:

1. *Query-type-aware routing*: send spear-fishing traffic to BM25 / Flat.
2. *Index-aware training*: penalize placement of rare features in poorly-reachable embedding regions.
3. *Hybrid retrieval*: sparse for precision on rare tokens, dense for semantic generalization—now with a quantitative justification.

Takeaways

1. Representation capacity is *necessary but not sufficient* for production dense retrieval.
2. ANN indices introduce a *second*, structurally different failure mode concentrated on rare-token / identifier queries.
3. We formalized this as **indexing bias** + **centroid-dilution bound**, then directly measured the centroid signal per query ($\sim 40\times$ collapse from standard to ID).
4. Reaching Flat-equivalent recall on ID requires 80–160$\times$ more `efSearch` budget than on standard—the corpus-invariant quantification of the ANN value-proposition collapse.
5. Supervised encoders don't escape this—they collapse at the representation stage on the same queries.

Code & experiments: `github.com/alemagnani/sparse-dense-paper`

Thank you

Questions?

Alessandro Magnani | ale.magnani@gmail.com
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